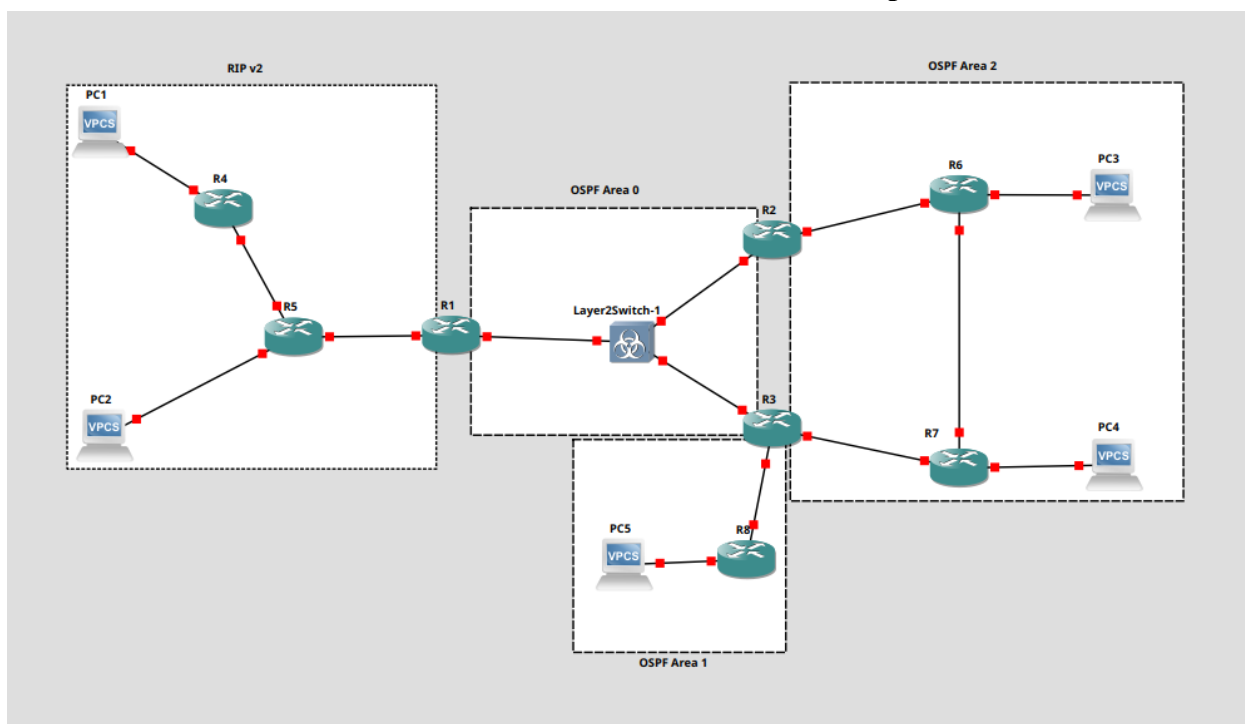


Отчет по лабораторной работе №5 модуля 4
Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF
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- Диапазон 192.168.x.x — для сегмента RIP v2.
- Диапазон 10.0.0.0/24 — для зоны OSPF Area 0 (мультидоступный сегмент через коммутатор Layer2Switch-1).
- Диапазон 10.1.x.x — для OSPF Area 1 (тупиковая зона).
- Диапазон 10.2.x.x — для OSPF Area 2.

Таблица подсетей и портов

Сегмент сети	Адрес подсети и примечания
Сеть PC1 – R4	192.168.14.0/24
Сеть PC2 – R5	192.168.15.0/24
Сеть R4 – R5	192.168.45.0/30 (маска /30 или 255.255.255.252 идеальна для стыков «точка-точка», так как содержит всего 2 полезных IP-адреса)
Сеть R5 – R1	192.168.51.0/30
Сеть Area 0 (R1 – R2 – R3)	10.0.0.0/24 (все три роутера подключены к одному коммутатору)
Сеть R3 – R8 (Area 1)	10.1.38.0/30
Сеть R8 – PC5 (Area 1)	10.1.85.0/24
Сеть R2 – R6 (Area 2)	10.2.26.0/30
Сеть R3 – R7 (Area 2)	10.2.37.0/30
Сеть R6 – R7 (Area 2)	10.2.67.0/30
Сеть R6 – PC3 (Area 2)	10.2.63.0/24
Сеть R7 – PC4 (Area 2)	10.2.74.0/24

PC1:

ip 192.168.14.10/24 192.168.14.1

save

PC2:

ip 192.168.15.10/24 192.168.15.1

save

PC3:

ip 10.2.63.10/24 10.2.63.1

save

PC4:

ip 10.2.74.10/24 10.2.74.1

save

PC5:

ip 10.1.85.10/24 10.1.85.1

save

R1:

enable

configure terminal

```
interface FastEthernet0/0
ip address 192.168.51.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.0.0.1 255.255.255.0
no shutdown
exit
Exit
Wr mem
```

```
Mar 1 00:02:51.891: %SYS-5-CONFIG-I-1: Configured from console by console
R1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.51.2	YES	manual	up	up
FastEthernet1/0	10.0.0.1	YES	manual	up	up

```
R1#
```

FastEthernet0/0 — 192.168.51.2/30

Подсеть: 192.168.51.0/30 (сегмент RIP v2)
Соединение: Линк между R1 и R5
Другой конец: R5 имеет адрес 192.168.51.1/30
Назначение: Пограничный интерфейс между RIP и OSPF (здесь настраивается редистрибуция)

FastEthernet1/0 — 10.0.0.1/24

Подсеть: 10.0.0.0/24 (OSPF Area 0)
Соединение: Подключен к коммутатору Layer2Switch-1
Соседи в сегменте: R2 (10.0.0.2) и R3 (10.0.0.3)
Назначение: Участие в OSPF Area 0, выборы DR/BDR

R2:

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.0.0.2 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.2.26.1 255.255.255.252
no shutdown
exit
```

R3:

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.0.0.3 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.2.37.1 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet2/0
ip address 10.1.38.1 255.255.255.252
no shutdown
exit
```

R4:

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 192.168.14.1 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 192.168.45.1 255.255.255.252
no shutdown
exit
```

R5:

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 192.168.15.1 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 192.168.45.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet2/0
ip address 192.168.51.1 255.255.255.252
no shutdown
exit
```

R6:

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.2.26.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet1/0
```

```
ip address 10.2.63.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
interface FastEthernet2/0
```

```
ip address 10.2.67.1 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
R7:
```

```
enable
```

```
configure terminal
```

```
interface FastEthernet0/0
```

```
ip address 10.2.37.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface FastEthernet1/0
```

```
ip address 10.2.74.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
interface FastEthernet2/0
```

```
ip address 10.2.67.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
R8:
```

enable

configure terminal

interface FastEthernet0/0

ip address 10.1.38.2 255.255.255.252

no shutdown

exit

interface FastEthernet1/0

ip address 10.1.85.1 255.255.255.0

no shutdown

exit

```
PC1> ping 192.168.14.1 -c 3

84 bytes from 192.168.14.1 icmp_seq=1 ttl=255 time=19.565 ms
84 bytes from 192.168.14.1 icmp_seq=2 ttl=255 time=10.780 ms
84 bytes from 192.168.14.1 icmp_seq=3 ttl=255 time=13.861 ms
```

С консоли PC1 пинг до его шлюза

```
*Mar  1 00:03:48.319: %SYS-5-CONFIG_I: Configured from console by console[OK]
R4#ping 192.168.45.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 36/56/64 ms
R4#
```

С консоли роутера R4 пинг до R5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/24 ms
R1#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/17/24 ms
R1#
```

С консоли роутера R1 доступность соседей в Area 0

Настройка R4:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.14.0
network 192.168.45.0
exit
```

Настройка R5:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.15.0
network 192.168.45.0
network 192.168.51.0
exit
```

Настройка R1:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.51.0
exit
```


show ip route rip

```
*Mar  1 00:28:50.983: %SYS-5-CONFIG_I: Configured from console by console[OK]
R4#show ip route rip
R    192.168.15.0/24 [120/1] via 192.168.45.2, 00:00:26, FastEthernet1/0
    192.168.51.0/30 is subnetted, 1 subnets
R        192.168.51.0 [120/1] via 192.168.45.2, 00:00:26, FastEthernet1/0
R4#
```

Таблица маршрутизации

```
PC1>
PC1> ping 192.168.15.10

192.168.15.10 icmp_seq=1 timeout
84 bytes from 192.168.15.10 icmp_seq=2 ttl=62 time=25.948 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=62 time=23.688 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=62 time=23.391 ms

```

PC1 пинг до PC2

```
PC1> ping 192.168.51.2

84 bytes from 192.168.51.2 icmp_seq=1 ttl=253 time=42.424 ms
84 bytes from 192.168.51.2 icmp_seq=2 ttl=253 time=40.528 ms
84 bytes from 192.168.51.2 icmp_seq=3 ttl=253 time=31.660 ms
84 bytes from 192.168.51.2 icmp_seq=4 ttl=253 time=24.935 ms
84 bytes from 192.168.51.2 icmp_seq=5 ttl=253 time=22.058 ms

PC1> 
```

PC1 пинг RIP-интерфейса роутера R1

Настройка R1:

configure terminal

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

exit

Настройка R2:

configure terminal

router ospf 1

```
network 10.0.0.0 0.0.0.255 area 0
network 10.2.26.0 0.0.0.3 area 2
exit
```

Настройка R3:

```
configure terminal
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 10.2.37.0 0.0.0.3 area 2
network 10.1.38.0 0.0.0.3 area 1
area 1 stub no-summary
exit
```

Настройка R8:

```
configure terminal
router ospf 1
network 10.1.38.0 0.0.0.3 area 1
network 10.1.85.0 0.0.0.255 area 1
area 1 stub
exit
```

Настройка R6:

```
configure terminal
router ospf 1
network 10.2.26.0 0.0.0.3 area 2
network 10.2.63.0 0.0.0.255 area 2
network 10.2.67.0 0.0.0.3 area 2
exit
```

Настройка R7:

```
configure terminal
```

```
router ospf 1
```

```
network 10.2.37.0 0.0.0.3 area 2
```

```
network 10.2.74.0 0.0.0.255 area 2
```

```
network 10.2.67.0 0.0.0.3 area 2
```

```
exit
```

```
show ip ospf neighbor
```

```
ne
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.2.26.1         1    FULL/DROTHER    00:00:37   10.0.0.2       FastEthernet0/0
192.168.51.2      1    FULL/DR         00:00:37   10.0.0.1       FastEthernet0/0
10.1.85.1         1    FULL/BDR        00:00:39   10.1.38.2      FastEthernet2/0
10.2.74.1         1    FULL/BDR        00:00:37   10.2.37.2      FastEthernet1/0
R3#
```

Проверка соседства

На рисунке показано, что маршрутизатор R3 (ABR) установил полнофункциональные соседские отношения (FULL) со всеми OSPF-соседями в трёх зонах: Area 0 (R1, R2), Area 1 (R8) и Area 2 (R7). Это подтверждает корректность настройки OSPF и готовность к обмену маршрутной информацией.

```
show ip route
```

```
R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.38.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.38.0/30 is directly connected, FastEthernet0/0
C       10.1.85.0/24 is directly connected, FastEthernet1/0
O*IA 0.0.0.0/0 [110/2] via 10.1.38.1, 00:02:22, FastEthernet0/0
R8#
```

Проверка Totally Stubby Area на R8

На рисунке представлена таблица маршрутизации R8 — внутреннего роутера Totally Stubby Area 1. Видно, что помимо двух напрямую подключенных сетей (10.1.38.0/30 и 10.1.85.0/24), в таблице присутствует единственный межзоновой маршрут — маршрут по умолчанию O*IA 0.0.0.0/0 через R3 (10.1.38.1). Отсутствие других маршрутов типа O IA подтверждает, что зона 1 настроена как полностью тупиковая (Totally Stubby): команда `area 1 stub no-summary` на ABR (R3) блокирует передачу внутрь зоны всех межзоновых маршрутов, заменяя их одним маршрутом по умолчанию. Это снижает размер таблицы маршрутизации и нагрузку на внутренние роутеры зоны.

ping 10.1.85.10

```
PC3> ping 10.1.85.10

10.1.85.10 icmp_seq=1 timeout
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=66.399 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=53.533 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=56.329 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=52.507 ms

PC3> █
```

Проверка пинга внутри зоны OSPF

Настройки на роутере R1:

configure terminal

router rip

redistribute ospf 1 metric 5

exit

router ospf 1

redistribute rip subnets

exit

Проверяем сторону RIP (на роутере R4):

show ip route

```

R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.14.0/24 is directly connected, FastEthernet0/0
R    192.168.15.0/24 [120/1] via 192.168.45.2, 00:00:10, FastEthernet1/0
    192.168.45.0/30 is subnetted, 1 subnets
C    192.168.45.0 is directly connected, FastEthernet1/0
    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
R    10.0.0.0/24 [120/6] via 192.168.45.2, 00:00:10, FastEthernet1/0
R    10.2.26.0/30 [120/6] via 192.168.45.2, 00:00:10, FastEthernet1/0
R    10.2.37.0/30 [120/6] via 192.168.45.2, 00:00:10, FastEthernet1/0
R    10.1.38.0/30 [120/6] via 192.168.45.2, 00:00:12, FastEthernet1/0
R    10.2.63.0/24 [120/6] via 192.168.45.2, 00:00:12, FastEthernet1/0
R    10.2.74.0/24 [120/6] via 192.168.45.2, 00:00:12, FastEthernet1/0
R    10.2.67.0/30 [120/6] via 192.168.45.2, 00:00:12, FastEthernet1/0
R    10.1.85.0/24 [120/6] via 192.168.45.2, 00:00:12, FastEthernet1/0
    192.168.51.0/30 is subnetted, 1 subnets
R    192.168.51.0 [120/1] via 192.168.45.2, 00:00:29, FastEthernet1/0
R4#
R4#

```

Сторона рип

Проверяем сторону OSPF (на роутере R6):

show ip route

```

R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E2 192.168.14.0/24 [110/20] via 10.2.26.1, 00:04:40, FastEthernet0/0
O E2 192.168.15.0/24 [110/20] via 10.2.26.1, 00:04:40, FastEthernet0/0
     192.168.45.0/30 is subnetted, 1 subnets
O E2   192.168.45.0 [110/20] via 10.2.26.1, 00:04:40, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O IA   10.0.0.0/24 [110/2] via 10.2.26.1, 00:04:40, FastEthernet0/0
C     10.2.26.0/30 is directly connected, FastEthernet0/0
O     10.2.37.0/30 [110/2] via 10.2.67.2, 00:04:40, FastEthernet2/0
O IA   10.1.38.0/30 [110/3] via 10.2.67.2, 00:04:41, FastEthernet2/0
       [110/3] via 10.2.26.1, 00:04:41, FastEthernet0/0
C     10.2.63.0/24 is directly connected, FastEthernet1/0
O     10.2.74.0/24 [110/2] via 10.2.67.2, 00:04:41, FastEthernet2/0
C     10.2.67.0/30 is directly connected, FastEthernet2/0
O IA   10.1.85.0/24 [110/4] via 10.2.67.2, 00:04:46, FastEthernet2/0
       [110/4] via 10.2.26.1, 00:04:46, FastEthernet0/0
     192.168.51.0/30 is subnetted, 1 subnets
O E2   192.168.51.0 [110/20] via 10.2.26.1, 00:04:46, FastEthernet0/0
R6#
R6#

```

Сторона оспф

Проверка тупиковой зоны (на роутере R8):

show ip route

```

R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.38.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.1.38.0/30 is directly connected, FastEthernet0/0
C     10.1.85.0/24 is directly connected, FastEthernet1/0
O*IA 0.0.0.0/0 [110/2] via 10.1.38.1, 00:06:51, FastEthernet0/0
R8#

```

Тупиковая зона проверка

ping 192.168.14.10

ping 192.168.15.10

ping 10.2.63.10

ping 10.2.74.10

ping 10.1.85.10

```
PC1> ping 192.168.14.10 c 3

192.168.14.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=5 ttl=64 time=0.001 ms

PC1> ping 192.168.15.10 c 3

192.168.15.10 icmp_seq=1 timeout
84 bytes from 192.168.15.10 icmp_seq=2 ttl=62 time=26.531 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=62 time=23.417 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=62 time=30.682 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=62 time=32.243 ms

PC1> ping 10.2.63.10 c 3

84 bytes from 10.2.63.10 icmp_seq=1 ttl=59 time=109.331 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=59 time=73.887 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=59 time=71.257 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=59 time=55.507 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=59 time=83.646 ms

PC1> ping 10.2.74.10 -c 3

10.2.74.10 icmp_seq=1 timeout
84 bytes from 10.2.74.10 icmp_seq=2 ttl=59 time=70.180 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=59 time=70.081 ms

PC1> ping 10.1.85.10 -c 3

84 bytes from 10.1.85.10 icmp_seq=1 ttl=59 time=75.422 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=59 time=74.981 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=59 time=86.252 ms

PC1> 
```

Пинги с ПК1

```
PC2> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=62 time=39.828 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=62 time=31.898 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=62 time=22.744 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=62 time=34.530 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=62 time=23.371 ms

PC2> ping 192.168.15.10

192.168.15.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=5 ttl=64 time=0.001 ms

PC2> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=60 time=66.434 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=60 time=54.394 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=60 time=70.011 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=60 time=62.455 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=60 time=64.949 ms

PC2> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=60 time=67.426 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=60 time=53.885 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=60 time=59.132 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=60 time=53.609 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=60 time=52.445 ms

PC2> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=60 time=72.334 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=58.220 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=52.315 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=43.725 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=71.623 ms

PC2> █
```

Пинги с ПК2


```
PC3> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=85.126 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=74.508 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=51.364 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=83.134 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=60.973 ms

PC3> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=61.179 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=51.858 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=73.459 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=62.048 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=61.190 ms

PC3> ping 10.2.63.10

10.2.63.10 icmp_seq=1 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=2 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=3 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=4 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=5 ttl=64 time=0.001 ms

PC3> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=62 time=36.399 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=62 time=40.256 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=62 time=23.985 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=62 time=32.374 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=62 time=27.840 ms

PC3> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=60 time=51.156 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=51.198 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=44.042 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=55.276 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=42.581 ms

PC3>
```

Пинги с ПК3

```
PC4> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=89.169 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=53.716 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=64.312 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=62.984 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=54.834 ms

PC4> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=70.850 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=72.942 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=64.549 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=55.455 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=67.105 ms

PC4> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=62 time=37.191 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=62 time=31.226 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=62 time=24.343 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=62 time=24.383 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=62 time=26.774 ms

PC4> ping 10.2.74.10

10.2.74.10 icmp_seq=1 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=2 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=3 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=4 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=5 ttl=64 time=0.001 ms

PC4> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=61 time=61.086 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=61 time=42.318 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=61 time=33.820 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=61 time=33.032 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=61 time=32.835 ms

PC4>
```

Пинги с ПК4

```
PC5> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=66.559 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=64.409 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=65.773 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=77.552 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=73.379 ms

PC5> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=51.524 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=66.011 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=64.275 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=44.802 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=54.942 ms

PC5> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=60 time=45.080 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=60 time=47.833 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=60 time=54.256 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=60 time=51.867 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=60 time=54.978 ms

PC5> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=61 time=62.201 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=61 time=43.584 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=61 time=43.723 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=61 time=54.494 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=61 time=33.907 ms

PC5> ping 10.1.85.10

10.1.85.10 icmp_seq=1 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=2 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=3 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=4 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=5 ttl=64 time=0.001 ms

PC5> █
```

Пинги с ПК5

No.	Time	Source	Destination	Protocol	Length	Info
11	26.869979	192.168.45.1	224.0.0.9	RIPv2	66	Response
17	55.165702	192.168.45.1	224.0.0.9	RIPv2	66	Response
19	66.508354	192.168.45.2	224.0.0.9	RIPv2	246	Response
23	87.540002	192.168.45.1	224.0.0.9	RIPv2	66	Response
30	119.516176	192.168.45.1	224.0.0.9	RIPv2	66	Response
33	133.097687	192.168.45.2	224.0.0.9	RIPv2	246	Response
36	151.530706	192.168.45.1	224.0.0.9	RIPv2	66	Response
42	181.238716	192.168.45.1	224.0.0.9	RIPv2	66	Response
46	202.269958	192.168.45.2	224.0.0.9	RIPv2	246	Response
48	211.952304	192.168.45.1	224.0.0.9	RIPv2	66	Response
54	242.052269	192.168.45.1	224.0.0.9	RIPv2	66	Response
60	269.712176	192.168.45.2	224.0.0.9	RIPv2	246	Response
61	278.886764	192.168.45.1	224.0.0.9	RIPv2	66	Response
67	301.418406	192.168.45.1	224.0.0.9	RIPv2	66	Response
72	333.209896	192.168.45.1	224.0.0.9	RIPv2	66	Response
75	341.199771	192.168.45.2	224.0.0.9	RIPv2	246	Response
79	363.767714	192.168.45.1	224.0.0.9	RIPv2	66	Response
84	394.493393	192.168.45.1	224.0.0.9	RIPv2	66	Response
88	414.775479	192.168.45.2	224.0.0.9	RIPv2	246	Response
92	424.759131	192.168.45.1	224.0.0.9	RIPv2	66	Response
97	457.476608	192.168.45.1	224.0.0.9	RIPv2	66	Response
101	481.863595	192.168.45.2	224.0.0.9	RIPv2	246	Response
103	485.621504	192.168.45.1	224.0.0.9	RIPv2	66	Response
109	514.645310	192.168.45.1	224.0.0.9	RIPv2	66	Response

- Frame 19: Packet, 246 bytes on wire (1968 bits), 246 bytes captured (1968 bits) on interface -, id 0 - Ethernet II, Src: cc:05:20:6f:00:10 (cc:05:20:6f:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09) - Internet Protocol Version 4, Src: 192.168.45.2, Dst: 224.0.0.9 - User Datagram Protocol, Port: 520, Dst Port: 520 - Routing Information Protocol - Command: Response (2) - Version: RIPv2 (2) - IP Address: 10.0.0.0, Metric: 6 - IP Address: 10.1.38.0, Metric: 6 - IP Address: 10.1.85.0, Metric: 6 - IP Address: 10.2.26.0, Metric: 6 - IP Address: 10.2.37.0, Metric: 6 - IP Address: 10.2.63.0, Metric: 6 - IP Address: 10.2.67.0, Metric: 6 - IP Address: 10.2.74.0, Metric: 6 - IP Address: 192.168.15.0, Metric: 1 - IP Address: 192.168.51.0, Metric: 1	<pre> 0000 01 00 5e 00 00 09 cc 05 20 6f 00 10 08 00 45 c0 ..A...o...E 0010 00 e8 00 00 00 02 11 e9 91 c0 a8 2d 02 e0 00 0020 00 09 02 08 02 00 d4 95 e4 02 02 00 00 00 02 0030 00 00 0a 00 00 ff ff ff 00 00 00 00 00 00 0040 00 06 00 02 00 0a 01 26 00 ff ff ff fc 00 00 &..... 0050 00 00 00 00 00 06 02 00 00 0a 01 55 00 ff ff U.... 0060 ff 00 00 00 00 00 00 00 00 02 02 00 00 0a 02 0070 1a 00 ff ff ff fc 00 00 00 00 00 00 00 02 0080 00 00 0a 02 25 00 ff ff ff fc 00 00 00 00 00 %..... 0090 00 06 02 00 0a 02 3f 00 ff ff ff 00 00 00 ?..... 00a0 00 00 00 00 00 02 00 00 0a 02 43 00 ff ff C.... 00b0 ff fc 00 00 00 00 00 00 00 02 02 00 0a 02 00c0 4a 00 ff ff ff 00 00 00 00 00 00 00 00 02 J..... 00d0 00 c0 a8 0f 00 ff ff ff 00 00 00 00 00 00 3..... 00e0 00 01 02 00 00 c0 a8 33 00 ff ff ff fc 00 00 00f0 00 00 00 00 01 </pre>
--	---

Между R4 и R5

Анализ пакета RIPv2

Метрики и параметры из пакета:

Протокол: RIPv2 (работает поверх UDP, порт источника/назначения: 520).

Тип сообщения (Command): Response (2) — пакет обновления (объявления) маршрутов.

Адрес отправителя (Source): 192.168.45.2 (интерфейс роутера R5).

Адрес назначения (Destination): 224.0.0.9 — групповой мультикаст-адрес протокола RIPv2.

Пример анонсируемой «внешней» сети (из OSPF):

IP Address: 10.1.85.0

Netmask: 255.255.255.0

Metric: 6

Пример анонсируемой «родной» сети (RIP):

IP Address: 192.168.15.0

Metric: 1

Описание происходящего:

На скриншоте зафиксирован процесс обмена маршрутной информацией RIPv2 на линке между R4 и R5. Маршрутизатор R5 отправляет пакет обновления Response на групповой адрес 224.0.0.9.

Внутри сообщения передается список известных ему подсетей с указанием их масок и метрик (количества хопов), что позволяет соседнему роутеру R4 актуализировать свою таблицу маршрутизации.

В блоке данных прослеживается разница между типами маршрутов. Внутренние сети сегмента RIP (например, 192.168.15.0) анонсируются с метрикой 1. В то же время импортированные из OSPF сети (например, 10.1.85.0) имеют метрику 6. Это подтверждает успешную настройку редистрибуции: к базовой метрике перераспределения 5, заданной на пограничном R1, прибавился один хоп при прохождении через R5 до R4.

No.	Time	Source	Destination	Protocol	Length	Info
4	2.499681	10.0.0.3	224.0.0.5	OSPF	98	Hello Packet
6	3.702351	10.0.0.2	224.0.0.5	OSPF	98	Hello Packet
8	5.990895	10.0.0.1	224.0.0.5	OSPF	98	Hello Packet
13	12.507565	10.0.0.3	224.0.0.5	OSPF	98	Hello Packet
14	13.679953	10.0.0.2	224.0.0.5	OSPF	98	Hello Packet
17	15.385144	10.0.0.1	224.0.0.5	OSPF	98	Hello Packet
24	22.485414	10.0.0.3	224.0.0.5	OSPF	98	Hello Packet
26	23.689066	10.0.0.2	224.0.0.5	OSPF	98	Hello Packet
27	25.382647	10.0.0.1	224.0.0.5	OSPF	98	Hello Packet
34	32.493941	10.0.0.3	224.0.0.5	OSPF	98	Hello Packet
35	33.682921	10.0.0.2	224.0.0.5	OSPF	98	Hello Packet
37	35.372320	10.0.0.1	224.0.0.5	OSPF	98	Hello Packet
43	42.484895	10.0.0.3	224.0.0.5	OSPF	98	Hello Packet
44	43.764563	10.0.0.2	224.0.0.5	OSPF	98	Hello Packet
46	45.397910	10.0.0.1	224.0.0.5	OSPF	98	Hello Packet


```

* Frame 8: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
* Ethernet II, Src: cc:01:1f:ff:00:10 (cc:01:1f:ff:00:10), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
* Internet Protocol Version 4, Src: 10.0.0.1, Dst: 224.0.0.5
* Open Shortest Path First
  * OSPF Header
  * OSPF Hello Packet
    Network Mask: 255.255.255.0
    Hello Interval [sec]: 10
    Options: 0x12, (L) LLS Data block, (E) External Routing
    Router Priority: 1
    Router Dead Interval [sec]: 40
    Designated Router: 10.0.0.1
    Backup Designated Router: 10.0.0.3
    Active Neighbor: 10.2.26.1
    Active Neighbor: 10.2.37.1
  * OSPF LLS Data Block
  
```

```

0000 01 00 5e 00 00 05 cc 01 1f ff 00 10 08 00 45 c0  ..A...Y.....E
0010 00 54 02 a8 00 00 01 59 cb e3 0a 00 00 01 e0 00  ..T...3.....
0020 00 05 02 01 00 34 c0 a8 33 02 00 00 00 00 91 e1  ....4...
0030 00 00 00 00 00 00 00 00 00 ff ff ff 00 00 0a  ....(.....
0040 12 01 00 00 09 28 0a 00 00 01 0a 00 03 0a 02  ....%.....
0050 1a 01 0a 02 25 01 ff f6 00 03 00 01 00 04 00 00  ....
0060 00 01
  
```

Между R2 и Layer2Switch-1

Анализ пакета OSPF

Метрики и параметры из пакета:

Протокол: OSPF (работает напрямую поверх IP, номер протокола 89).

Тип сообщения (OSPF Packet Type): Hello Packet (1) — служебный пакет для установления и поддержания соседства.

Адрес отправителя (Source): 10.0.0.1 (интерфейс роутера R1).

Адрес назначения (Destination): 224.0.0.5 — групповой мультикаст-адрес всех OSPF-маршрутизаторов.

Network Mask: 255.255.255.0 (маска подсети интерфейса).

Hello Interval: 10 секунд — период отправки Hello-пакетов.

Router Dead Interval: 40 секунд — время ожидания перед объявлением соседа недоступным.

Designated Router (DR): 10.0.0.1 (выбранный главный маршрутизатор сегмента).

Backup Designated Router (BDR): 10.0.0.3 (резервный главный маршрутизатор).

Active Neighbors: 10.2.26.1 (сосед R2) и 10.2.37.1 (сосед R3).

Описание происходящего:

На скриншоте представлен анализ периодического пакета OSPF Hello в многоадресном сегменте Area 0 (сеть коммутатора Layer2Switch-1). Маршрутизатор R1 транслирует данный пакет на мультикаст-адрес 224.0.0.5. Hello-пакеты отправляются каждые 10 секунд, а таймер удержания составляет 40 секунд.

Данные пакета подтверждают, что в общем сегменте успешно завершился процесс выборов выделенных маршрутизаторов: роль DR занял R1 (10.0.0.1), а роль BDR — R3 (10.0.0.3). В поле Active Neighbor зафиксировано успешное обнаружение соседних устройств R2 (10.2.26.1) и R3 (10.2.37.1). Идентичность параметров сетевой маски и таймеров во всех Hello-пакетах подтверждает отсутствие конфигурационных ошибок на стыке, что позволило роутерам перейти в состояние смежности FULL.